MILITARY SPECIFICATION

TURNTABLE, AIRCRAFT, COMPASS COMPENSATING, TYPE C

This specification has been approved
by the Bureau of Aeronautics, Department of the Navy

1. SCOPE

1.1 Scope: This specification covers a type of compass compensating turntable, designated Type C.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards and bulletins, of the issue in effect on the date of invitation for bids, shall form a part of this specification to the extent specified herein:

SPECIFICATIONS:

Federal:

QQ-A-318	Aluminum Alloy Plate and Sheet 5052

Military:

MIL-C-21180	Castings, Aluminum Alloy, High Strength
MIL-C-132	Crate, Wood, Open; Maximum Capacity 2500 Pounds
MIL-D-5028	Drawings and Data Lists; Preparation of Manufactures' (For Production, Aeronautical and Associated Equipment)
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, (General Specification For)
MIL-A-8625	Anodic Coatings, For Aluminum and Aluminum Alloys
MIL-M-7911	Marking, Identification of Aeronautical Equipment, Assemblies, and Parts
MIL-C-5541	Chemical Films For Aluminum and Aluminum Alloys

STANDARDS:

MIL-STD-129	Marking For Shipment and Storage

PUBLICATIONS:

Air Force-Navy Aeronautical Bulletins

No. 143	Specifications And Standards; Use of

(Copies of this specification and applicable specifications and bulletins may be obtained upon application to the Commanding Officer, Naval Aviation Supply Depot, 700 Robbins Avenue, Philadelphia 11, Pennsylvania, Attention Code AD).

3. REQUIREMENTS

3.1 Component Parts: The turntable shall consist of an upper table, lower table, hinged flap, and bearing.

3.2 Material:

3.2.1 Specifications and Standards: Specifications and standards for all materials, parts, and Government certification and approval of processes and equipment, which are not specifically designated herein, and which are necessary for the execution of this specification, shall be selected in accordance with Bulletin No. 143, except as provided in the following paragraph:

FSC 1730

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3.2.1.1 Standard Parts: AN or MIL standard parts shall be used wherever they are suitable for the purpose, and shall be identified by their part numbers. Commercial utility parts such as screws, bolts, nuts, cotter pins, et cetera, may be used, provided they have suitable properties and are replaceable by the AN or MIL standard parts without alteration, and provided the corresponding AN or MIL part numbers are referenced on the drawings and in the parts lists. In applications for which no suitable corresponding AN or MIL part is in effect on date of invitation for bids, commercial parts may be used provided they conform to the requirements of this specification.

3.2.2 Protective Treatment: When materials are used in the construction of the turntable that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.2.3 Metal Parts: All metal parts used in the construction of the turntable shall be of nonmagnetic material and shall conform to either Federal specification QQ-A-318 (temper H112) or Military specification MIL-C-21180 (Alloy No.. A-356 and Grade C soundness).

3.3 Design: The turntable shall conform substantially to Figures 1 and 2.

3.3.1 Lower Part: The lower part of the turntable shall be designed to be set in concrete. It shall be flanged on its upper surface to prevent water from running into the bearing.

3.3.2 Upper Part: The upper part of the turntable shall rest upon the lower part of the base and rotate freely. The upper surface shall be flat so that a dual airplane wheel can be taxied onto and off of the turntable easily and without damage to any enclosures, cowlings, or any parts of the landing gear. A hinged flap shall be provided, substantially as shown on Figures 1 and 2. It shall be of sufficient size and substantial enough to act as a wheel stop when in the raised position and shall be designed so it may be lowered to permit taxiing the airplane off of the turntable.

3.4 Construction: The turntable shall be constructed so that no parts will work loose in service. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service.

3.4.1 Bearing: The contact area between the upper and lower parts shall be sufficient to support the load applied without seizing or galling. A low friction material may be interposed between the bearing surfaces, if desired. If a lubricant is used, it shall withstand a temperature of $-70^{\circ}C$ without congealing and $+50^{\circ}C$ without evaporating. The lubricant shall not attack any of the parts.

3.4.2 Load Capacity: The entire assembly shall be constructed to support and operate under a load of 150,000 pounds with the load applied as a dual airplane wheel. It shall support such a load and operate without binding due to flexing of parts.

3.4.3 Azimuth Scale And Markings: The lower part of the turntable shall have a circular flange around the outer edge upon which there is a 360-degree azimuth scale. This scale shall be divided into 1-degree graduations. The 10 and 5-degree graduations shall be raised "V" in cross section. The 10-degree graduations shall extend the full width of the flange. The 5-degree graduations shall be half the length of the 10-degree graduations. The 1-degree graduations shall be the same length as the 5-degree graduations, but shall be engraved "V" section instead of raised. Each graduation shall be accurate in azimuth to within $\pm 1/4$ degree. The cardinal points on the azimuth scale shall be marked with the letters N, E, S, and W. The graduations at 30-degree intervals shall be marked 3, 6, 9, 12, et cetera. The letters and figures shall be not less than 1/2 inch high and of the same raised "V" sections as the 10 and 5-degree graduations.

3.4.4 Index Pointer: There shall be an index pointer on the rotating part of the turntable for indicating on the azimuth scale on the lower part of the turntable. It shall be readily adjustable without the use of tools through a range of 60 degrees.

3.4.5 Remote Indication: Provisions shall be incorporated in the turntable for a transmitter for remote indication of the heading of the airplane. Means shall be provided for synchronizing the transmitter and the index pointer which reads on the azimuth scale.

3.4.6 Cover: A suitable cover shall be provided for the turntable to protect it from the elements when it is not in use. A cover shall be substantial enough to withstand service

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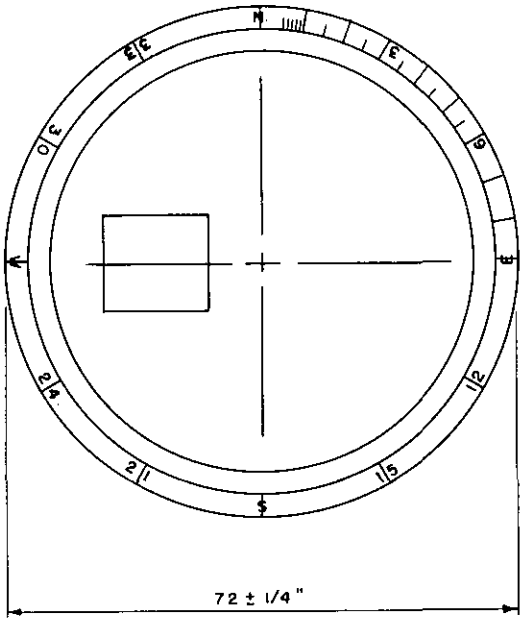


FIG. 1 PLAN VIEW

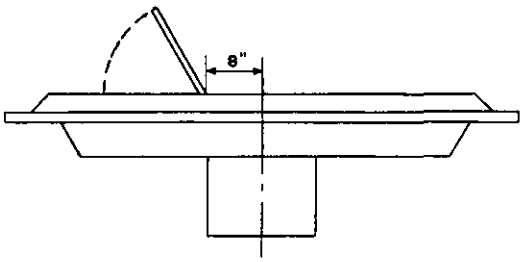


FIG. 2 ELEVATION

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usage and shall be fitted with suitable handles for removing and replacing. It shall be of nonmagnetic material throughout and, if of metal, shall be given the same protective coating and finish as the turntable.

3.5 Interchangeability: All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance. Changes in manufacturer's part numbers shall be governed by the drawing number requirements of Specification MIL-D-5028.

3.6 Weight: The weight of the complete assembly shall not exceed 600 pounds.

3.7 Finishes And Protective Coatings:

3.7.1 Aluminum Alloy Parts: Aluminum alloy parts shall be covered with an anodic film conforming to Specification MIL-A-8625, except as follows:

3.7.1.1 Aluminum alloys which do not anodize satisfactorily shall be coated with a chemical film in accordance with Specification MIL-C-5541.

3.8 Nameplate: A nameplate, permanently and legibly filled in with the following information, shall be securely attached to the turntable:

Turntable, Aircraft, Compass Compensating
Type C
Specification MIL-T-21596(Aer)
(Mfr's. Name or Trademark)
U. S. Property
Mfr's. Part No. _____

3.8.1 Manufacturer's Part Number: The manufacturer's part number marked in the space provided on the nameplate shall be in accordance with Specification MIL-M-7911. The part number shall be a concise identifying number and shall not be used to identify any other item.

3.9 Workmanship:

3.9.1 General: The turntable, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to soundness of any castings used, sharpness and legibility of all cast markings and engravings, smoothness of finish, and freedom of parts from burrs and dangerously sharp machined edges.

3.9.2 Dimensions: Dimensions and tolerances not specified, shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the turntable, they shall be held or limited accordingly.

4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of Tests: The inspection and testing of turntables shall be classified as follows:

4.1.1 Inspection Tests: Inspection tests are those tests accomplished on turntables submitted for acceptance under contract.

4.1.2 Preproduction Tests: Preproduction tests are those tests accomplished on a sample representative of the production turntable to determine that it meets the requirement of this specification.

4.1.3 Unless otherwise specified, the preproduction model shall be subject to all tests listed under 4.2 and 4.3.

4.2 Test Conditions:

4.2.1 Standard Atmospheric Conditions: Whenever the pressure and temperature existing at the time of the test are not specified definitely, it is understood that the test is to be made at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately +25°C). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the difference from the specified condition.

4.2.2 Attitude: Unless otherwise specified, the turntable shall be tested in the normal installed position.

4.3 Inspection Tests:

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4.3.1 Unless otherwise specified, contractor's records of all inspection work and tests, giving the results of tests required to determine compliance with the requirements and tests specified herein, shall be kept complete and shall be available to the Government representative at all times. The record or report of inspection and tests shall be signed or approved by a responsible person specifically assigned by the Contractor. Contractors not having laboratory testing facilities satisfactory to the Government shall engage the services of a commercial testing laboratory capable of conducting tests to determine compliance with all the requirements and tests in the specification and acceptable to the Government.

4.3.2 Individual Tests: Each turntable shall be subjected to the following tests:

4.3.2.1 Examination of Product: Each turntable shall be inspected to determine compliance with the requirements specified herein with respect to material, workmanship, finishes, scale accuracy, and marking.

4.3.2.2 Functional: Each turntable shall be operated to determine that the component parts fit properly and operate freely without shake, wobble, or binding.

4.3.3 Sampling Tests: At least one turntable shall be selected at random from each lot of the order and subjected to the following tests:

(A lot shall consist of turntables manufactured under essentially the same conditions and submitted for inspection at substantially the same time.)

4.3.3.1 When tests are specified on a quantity of turntables that are selected as representative of a certain lot, and one or more of this number fails to meet the specified tests, additional turntables of the lot represented shall be tested immediately to determine the extent of failure. Individual performance tests shall not be interrupted, unless the defect is of such a nature that it will seriously affect the performance or safe use of the turntable.

4.3.3.2 Low Temperature: After exposure in a cold chamber at -70° ±2°C for a period of 24 hours, the turntable shall operate without excessive drag from the effect of congealed lubricant or contraction of metals.

4.3.3.3 High Temperature: After exposure to a temperature of +50°C for a period of 24 hours, the turntable shall operate without excessive drag from the effect of evaporating lubricant or expanding metals.

4.3.3.4 Salt Spray: The salt spray test shall be in accordance with salt spray tests, Procedure I of Specification MIL-E-5272, for a period of 50 hours. Upon completion of the test, the turntable shall be in operable condition.

4.3.3.5 Sand and Dust: The sand and dust test shall be in accordance with sand and dust tests Procedure I of Specification MIL-E-5272, with the cover in place. Upon completion of the test, the turntable shall be in operable condition.

4.4 All parts, specimens, or assemblies destroyed in making tests required by this specification and/or drawings, to determine compliance with the specification and/or drawings, shall be in addition to the quantity specified in the contract or purchase order and shall be furnished without increasing the cost of the contract or order.

5. PREPARATION FOR DELIVERY

5.1 General: The requirements specified herein apply only to direct purchases by or direct shipment to the Government.

5.2 Domestic Packing: Unless otherwise specified, each turntable shall be packed for domestic shipment in a container constructed to insure acceptance by common or other carrier, for safe transportation, at the lowest rate, to the point of delivery. Except as specified herein, the container shall conform to Specification MIL-C-132 and the requirements of the Consolidated Freight Classification Rules in effect at the time of shipment. The container shall be capable of withstanding storage, rehandling, and reshipment without the necessity of repacking, and may be expendable.

5.3 Marking:

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5.3.1 Exterior Shipping Container: Each exterior shipping container shall be durably and legibly marked in accordance with MIL-STD-129 with the following information in such a manner that the markings will not become damaged when the package is opened:

Turntable, Aircraft, Compass Compensating
Specification MIL-T-21596(Aer)
Type C
Mfr's. Part No. _____
Mfr's. Serial No. _____
Order No. _____
Stock No. _____
Date of Manufacture _____
Quantity _____
(*Mfr's. Name or Trademark*)
Name of Contractor (if not the same as manufacturer)

6. NOTES

6.1 Ordering Data: Procurement documents should specify the title, number, and date of this specification.

6.2 Use: The Type C turntable covered by this specification is intended for use in swinging compasses on airplanes which have single or multiple wheels on each main landing gear.

NOTICE.- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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